Foundations of Generalised Linear Models, v10

From Conditional Expectation to Logistic Regression

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Technical notes, 2026

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1 Two approaches to the regression function

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(X, Y) : \Omega \to \mathbb{R}^2$ a random vector with $\mathbb{E}|Y| < \infty$.

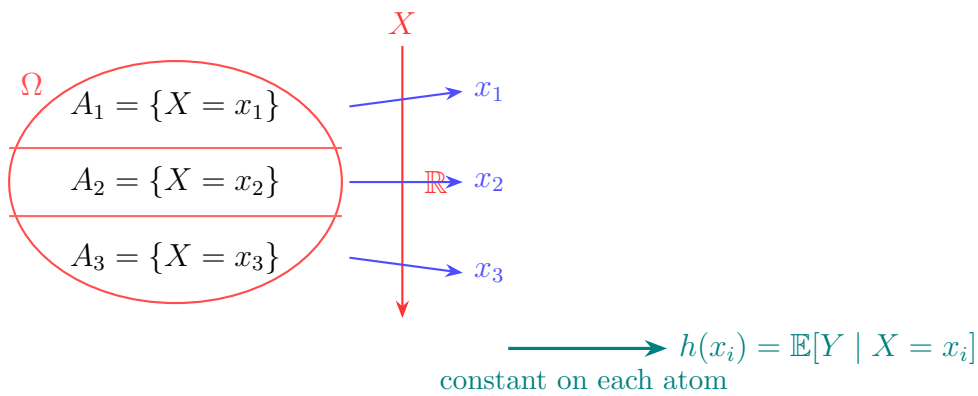
1.1 Motivation: the discrete case

Suppose X takes values $x_1, x_2, \ldots$ with $\mathbb{P}(X = x_i) > 0$. The sets $A_i = \{X = x_i\}$ are the *atoms* of $\sigma(X)$.

Lemma 1.1 (Measurable functions are constant on atoms)**.** Let $A \in \mathcal{S}$ be an atom of a sub-σ-algebra $\mathcal{S} \subseteq \mathcal{F}$. If $f : \Omega \to \mathbb{R}$ is $\mathcal{S}$-measurable, then f is $\mathbb{P}$-a.e. constant on A.

Proof. For any t, $\{f \le t\} \cap A \in \mathcal{S}$. Since A is an atom, its $\mathbb{P}$-measure is either 0 or $\mathbb{P}(A)$. Hence the conditional distribution of f on A is a point mass, i.e. $f = c$ $\mathbb{P}$-a.e. on A for some $c \in \mathbb{R}$. □

Remark 1.1. For $\sigma(X)$ the atoms are the level sets $A_i = \{X = x_i\}$. Lemma 1.1 says $\tilde{h}(X(\cdot))$ is constant on each A_i, so $h(x_i) = \mathbb{E}[Y \mid X = x_i]$ is a well-defined single number.



Elementary definition: $h(x_i) := \mathbb{E}[Y \mid X = x_i] = \mathbb{E}[Y \cdot \mathbf{1}_{A_i}]/\mathbb{P}(A_i)$, giving the key property $\mathbb{E}[Y \cdot \mathbf{1}_A] = \mathbb{E}[h(X) \cdot \mathbf{1}_A]$ for all $A \in \sigma(X)$. (**??**) For continuous X this becomes the general definition via R–N.

1.2 The Radon–Nikodym regression function

Remark 1.2 (Three-step construction)**. Step 1. Use Y as a density to define ν.** $\nu(B) := \int_{\{X \in B\}} Y \, d\mathbb{P}$.

Step 2. Move to Range(X) **with pushforward $\mathbb{P}_X$.** The essential domain of h is $\mathrm{supp}(\mathbb{P}_X) \subseteq \mathbb{R}$; values of h outside it are irrelevant.

Step 3. R–N density: $h = d\nu/d\mathbb{P}_X$.

$\mathbb{P}$ need not be a probability measure; any σ-finite measure works.

$$\boxed{\begin{array}{c}\text{Step 1}\\ \nu(B)=\int_{\{X\in B\}} Y\,d\mathbb{P}\end{array}} \xrightarrow{\text{pushforward}} \boxed{\begin{array}{c}\text{Step 2}\\ \text{space Range}(X),\ \text{measure } \mathbb{P}_X\end{array}} \xrightarrow{\text{R–N thm}} \boxed{\begin{array}{c}\text{Step 3}\\ h=d\nu/d\mathbb{P}_X\end{array}}$$

"Y-mass per unit X-probability"

Definition 1.1 (R–N regression function of Y on X w.r.t. $\mathbb{P}$)**.** $h:\mathbb{R}\to\mathbb{R}$ is the *R–N regression function of Y on X w.r.t.* $\mathbb{P}$ if

$$\int_{\{X\in B\}} Y\,d\mathbb{P}=\int_B h\,d\mathbb{P}_X \qquad \forall\, B\in\mathcal{B}(\mathbb{R}),$$

i.e. $h=d\nu/d\mathbb{P}_X$ where $\nu(B):=\int_{\{X\in B\}} Y\,d\mathbb{P}$.

Remark 1.3 (Full name: regression of Y on X w.r.t. $\mathbb{P}$)**. Of** Y: Y is being averaged. **On** X: conditioning on the value of X; h lives on Range(X). **W.r.t.** $\mathbb{P}$: $\mathbb{P}$ is the weighting measure; changing $\mathbb{P}$ changes h even if (X,Y) as functions are fixed. In short: *$h(x)$ is the $\mathbb{P}$-average of Y over $\{\omega : X(\omega)=x\}$.*

Remark 1.4 (Difficulties with signed measures and why $\mathbb{E}|Y|<\infty$ resolves them)**.** A natural question arises of whether there are difficulties in defining a Radon–Nikodym derivative for a signed measure. The answer is: yes, there are, and the assumption $\mathbb{E}|Y|<\infty$ is precisely what disposes of them.

What is a signed measure? A signed measure on a measurable space $(\mathcal{X},\mathcal{A})$ is a countably additive set function $\nu:\mathcal{A}\to[-\infty,+\infty]$ that takes at most one of the values $\pm\infty$. The restriction "at most one" is necessary to avoid the indeterminate form $\infty-\infty$ in the additivity condition. By the *Hahn decomposition theorem*, $\mathcal{X}$ splits into a positive set P and negative set N (where ν is non-negative and non-positive respectively), and one writes $\nu=\nu^+-\nu^-$ where $\nu^+(B)=\nu(B\cap P)$ and $\nu^-(B)=-\nu(B\cap N)$ are positive measures. This is the *Jordan decomposition*, unique up to $|\nu|:=\nu^++\nu^-$ (the total variation measure).

The R–N theorem for signed measures and its difficulty. If ν is a signed measure and μ is a σ-finite positive measure with $\nu\ll\mu$, one can formally write $h=d\nu^+/d\mu-d\nu^-/d\mu$. The difficulty is *integrability*: if $\int h^+\,d\mu=+\infty$ and $\int h^-\,d\mu=+\infty$ simultaneously, then for some sets B the integral $\int_B h\,d\mu$ is of the form $\infty-\infty$ and is undefined. The standard resolution requires at least one of $\nu^+(\mathcal{X})$ or $\nu^-(\mathcal{X})$ to be finite — i.e. ν must be a genuine signed measure, not merely a difference of two infinite positive measures.

Why there is no difficulty in our setting. Our signed measure is $\nu(B)=\int_{\{X\in B\}} Y\,d\mathbb{P}$, with Jordan decomposition $\nu^+(B)=\int_{\{X\in B\}} Y^+\,d\mathbb{P}$ and $\nu^-(B)=\int_{\{X\in B\}} Y^-\,d\mathbb{P}$. The condition $\mathbb{E}|Y|<\infty$ means:

$$\nu^+(\mathbb{R})=\mathbb{E}Y^+=\int Y^+\,d\mathbb{P}<\infty,\qquad \nu^-(\mathbb{R})=\mathbb{E}Y^-=\int Y^-\,d\mathbb{P}<\infty.$$

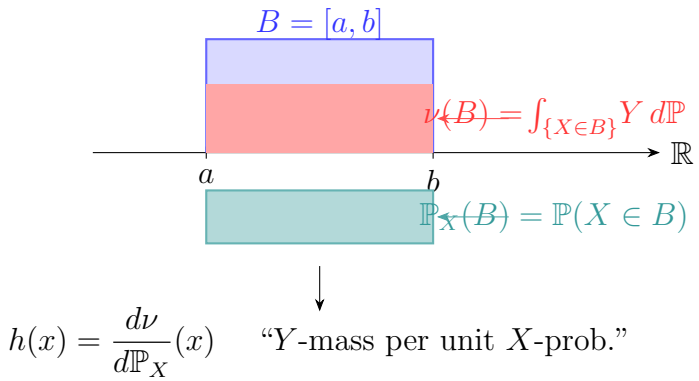
Both are finite, so ν is a *finite* signed measure, and the R–N derivative $h=d\nu/d\mathbb{P}_X$ is a $\mathbb{P}_X$-integrable function with no $\infty-\infty$ ambiguity. The assumption $\mathbb{E}|Y|<\infty$ is not a technicality: it is the exact condition needed to make the signed measure ν well-posed.

References.

- P. Halmos, *Measure Theory* (1950), Chapter VI: classical treatment of Hahn and Jordan decompositions.
- W. Rudin, *Real and Complex Analysis*, 3rd ed. (1987), Theorem 6.12: R–N theorem for complex (hence signed) measures; finiteness conditions discussed explicitly.
- V. Bogachev, *Measure Theory*, Vol. 1 (2007), §3.10: modern treatment including the σ-finite case and subtleties of the signed setting.
- N. Dunford & J.T. Schwartz, *Linear Operators*, Vol. 1 (1958), III.7: signed measures as elements of the dual of C_0; functional-analytic perspective.

Theorem 1.1 (Existence, uniqueness, properties)**.** Under $\mathbb{E}|Y| < \infty$: h exists; unique $\mathbb{P}_X$-a.e.; essential domain $\operatorname{supp}(\mathbb{P}_X)$; $\int h\, d\mathbb{P}_X = \mathbb{E}Y$; $h \in [c, d]$ a.e. when $Y \in [c, d]$ a.s.

Remark 1.5 (Density analogy)**.** h is an $L^1(\mathbb{P}_X)$-equivalence class, free on null sets, integrating to $\mathbb{E}Y$ — exactly as probability densities integrate to 1.



1.3 The conditional-expectation regression function

Definition 1.2 (Abstract conditional expectation)**.** $\mathbb{E}(Y \mid \mathcal{G})$ is the unique (up to $\mathbb{P}$-a.s.) $\mathcal{G}$-measurable r.v. with $\int_G \mathbb{E}(Y|\mathcal{G})\, d\mathbb{P} = \int_G Y\, d\mathbb{P}$ for all $G \in \mathcal{G}$.

Definition 1.3 (C.E. regression function of Y on X w.r.t. $\mathbb{P}$)**.** Set $Z := \mathbb{E}(Y \mid \sigma(X))$. By Doob–Dynkin, $Z = \tilde{h}(X)$ $\mathbb{P}$-a.s. for some Borel $\tilde{h}$. Any such $\tilde{h}$ is the *C.E. regression function of Y on X w.r.t. $\mathbb{P}$.*

Remark 1.6 (Role of $\mathbb{P}$: R–N vs. C.E.)**.**

R–N	**C.E.**
$\mathbb{P}$ builds $(\nu, \mathbb{P}_X)$, then steps aside. h lives on $\mathbb{R}$.	$\mathbb{P}$ used throughout on Ω. $\tilde{h}$ derived via Doob–Dynkin.
Less probabilistic in spirit.	Fundamentally probabilistic.
Works for any σ-finite $\mathbb{P}$.	Needs probability structure.

Remark 1.7 (Etymology)**.** "Regression" from Galton (1886). Abstract C.E. formalised by Kolmogorov (1933).

1.4 Equivalence

Theorem 1.2 ($h = \tilde{h}$ $\mathbb{P}_X$-a.e.)**.** Any R–N and any C.E. regression function agree $\mathbb{P}_X$-a.e.

Proof. For any B: $\int_B \tilde{h}\, d\mathbb{P}_X = \int_{\{X\in B\}} Z\, d\mathbb{P} = \int_{\{X\in B\}} Y\, d\mathbb{P} = \nu(B)$. So $\tilde{h}$ satisfies Def. 1.1; uniqueness gives $\tilde{h} = h$ $\mathbb{P}_X$-a.e. □

Remark 1.8 (Notation)**.** $\mathbb{E}(Y \mid X = x) := h(x)$ is shorthand only.

2 Optimality and the linear projection

Theorem 2.1 (h is the L^2-best predictor)**.** Let $Z := \mathbb{E}(Y \mid X)$ and U be any r.v. with $U \in \sigma(X)$ and $\mathbb{E}U^2 < \infty$. Then

$$\mathbb{E}(Y - Z)^2 \;\le\; \mathbb{E}(Y - U)^2.$$

Equality holds iff $U = Z$ $\mathbb{P}$-a.s.

Proof . Start with the RHS, adding and subtracting Z:

$$\begin{aligned}\mathbb{E}(Y-U)^2 &= \mathbb{E}\big[(Y-Z)+(Z-U)\big]^2\\ &= \underbrace{\mathbb{E}(Y-Z)^2}_{=:\,a} + \underbrace{2\,\mathbb{E}\big[(Y-Z)(Z-U)\big]}_{=:\,2b} + \underbrace{\mathbb{E}(Z-U)^2}_{=:\,c}.\end{aligned}$$

Step 1: $b = 0$. Since $Z - U$ is $\sigma(X)$-measurable, pull it out of the inner conditional expectation using the tower property:

$$\begin{aligned}\mathbb{E}\big[(Y-Z)(Z-U)\big] &= \mathbb{E}\Big[\mathbb{E}\big[(Y-Z)(Z-U) \mid X\big]\Big]\\ &= \mathbb{E}\Big[(Z-U)\,\mathbb{E}\big[Y-Z \mid X\big]\Big].\end{aligned}$$

Now $\mathbb{E}[Y - Z \mid X] = \mathbb{E}[Y \mid X] - Z = Z - Z = 0$. Therefore $b = 0$.

Step 2: $c \ge 0$. $c = \mathbb{E}(Z - U)^2 \ge 0$ by non-negativity of the square.

Combining: $\mathbb{E}(Y-U)^2 = \mathbb{E}(Y-Z)^2 + \mathbb{E}(Z-U)^2 \ge \mathbb{E}(Y-Z)^2$. Equality iff $\mathbb{E}(Z-U)^2 = 0$, i.e. $U = Z$ $\mathbb{P}$-a.s. □

Remark 2.1 (Non-uniqueness as a pointwise function)**.** The minimiser is not unique as a pointwise function: any U that equals Z $\mathbb{P}$-a.s. also achieves the minimum. The minimiser is unique as a $\mathbb{P}$-a.s. equivalence class, or equivalently as an element of $L^2(\mathbb{P})$ (the non-uniqueness argument from v2 notes). The key identity in the proof is $\mathbb{E}[Y - Z \mid X] = 0$: this is why the cross-term b vanishes, and this is the structural role of the conditional expectation.

3 The parabola example

3.1 Setup: a degenerate but instructive case

The key observation: the parabola example is the most degenerate possible instance of the R–N construction.

Take:

$$(\Omega, \mathcal{F}, \mathbb{P}) = \big([0,1],\, \mathcal{B}([0,1]),\, \mathrm{Leb}\big), \quad \omega \in [0,1] \subseteq \mathbb{R}.$$

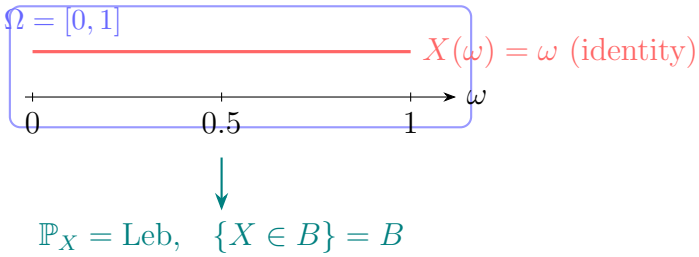
Define:

$$X(\omega) = \omega \quad \text{(identity map)}, \qquad Y(\omega) = (X(\omega))^2 = \omega^2 = h(X(\omega)).$$

Two key degeneracies follow immediately:

(1) $X(\omega) = \mathrm{id}$, so the event $\{X \in B\} = \{\omega : X(\omega) \in B\} = B$.

(2) $\mathbb{P} = \mathrm{Leb}$ and $X = \mathrm{id}$, so the pushforward $\mathbb{P}_X = \mathrm{Leb}$ as well: for any B, $\mathbb{P}_X(B) = \mathbb{P}(X \in B) = \mathbb{P}(\omega \in B) = \mathrm{Leb}(B)$. Thus $\mathbb{P} = \mathbb{P}_X$ in this example.



3.2 The R–N construction collapses to a trivial density

With $\{X \in B\} = B$ and $\mathbb{P} = \mathrm{Leb}$:

$$\nu(B) = \int_{\{X \in B\}} Y \, d\mathbb{P} = \int_B \omega^2 \, d\omega = \int_B x^2 \, dx.$$

Reference measure: $\mathbb{P}_X = \mathrm{Leb}$. R–N density: $h = d\nu/d\mathrm{Leb} = x^2$.

There is no averaging taking place: Y is a deterministic function of X (in fact $Y = h \circ X$ by construction), so h simply reads off the value of Y at each x. The regression function is trivial precisely because there is no randomness in Y given X.

Remark 3.1 (Why this is the degenerate case)**.** In a general regression problem, Y is not determined by X: for the same x there is a distribution of Y-values, and $h(x)$ averages them. Here the "distribution of Y given $X = x$" is the point mass at x^2, so the average is x^2 itself.

This is made precise as follows: since $X = \mathrm{id}$ and $\mathbb{P} = \mathbb{P}_X$, the three-step construction (Remark 1.2) degenerates:

- $\nu(B) = \int_B x^2 \, dx$ (direct integral, no "restriction" needed),
- $\mathbb{P}_X = \mathrm{Leb} = \mathbb{P}$ (no pushforward needed),
- $h = x^2$ (density of $x^2 \, dx$ w.r.t. dx, i.e. just x^2).

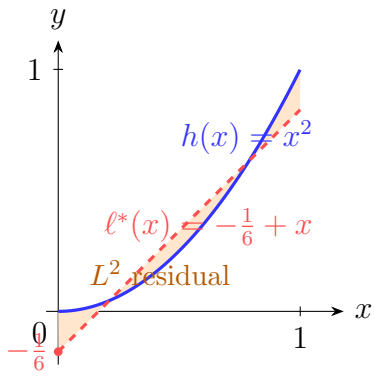
3.3 The affine projection ℓ^*

Minimise $f(a, b) = \int_0^1 (x^2 - (a + bx))^2 \, dx$:

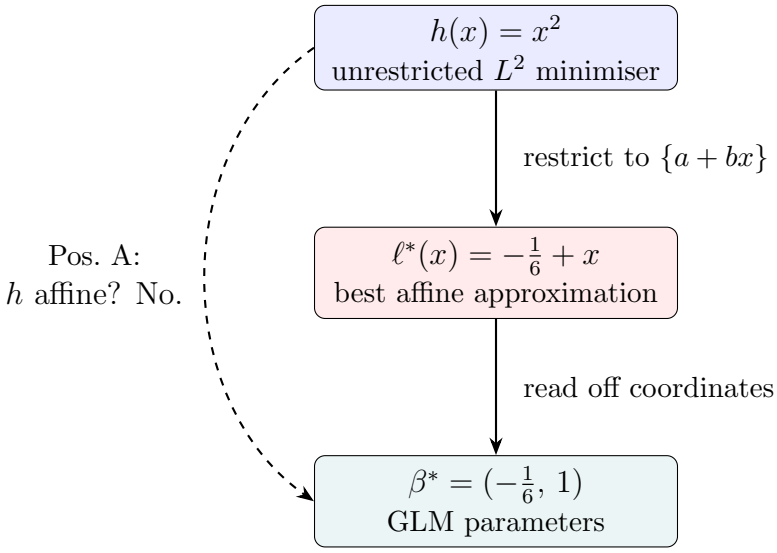
$$f(a, b) = \tfrac{1}{5} - \tfrac{2a}{3} - \tfrac{b}{2} + a^2 + ab + \tfrac{b^2}{3}.$$

(Coefficient of b is $-\frac{1}{2}$: the notes had a factor-of-2 error.) First-order conditions give $a = -\frac{1}{6}$, $b = 1$:

$$\boxed{\ell^*(x) = -\tfrac{1}{6} + x.}$$



3.4 Three objects and the GLM



Remark 3.2 (Positions A and B)**.** **A**: requires $h = x^2$ to be affine — false here. **B**: $\beta^* = (-\frac{1}{6}, 1)$ are best-approximation parameters; estimating them is meaningful even under misspecification.

4 The underspecification argument

Remark 4.1 (Free g is vacuous; fixing g is substantive)**.** If h injective: set $g = h^{-1}$, $\beta_0 = 0$, $\beta_1 = 1$. Any injective h satisfies the GLM equation. Fixing g is a genuine modelling commitment.

Assumption 4.1 (Position A)**.** $h(x) = g^{-1}(\beta_0^* + \beta_1^* x)$ $\mathbb{P}_X$-a.e. MLE consistent and asymptotically normal.

Remark 4.2 (Position B)**.** $\beta^* = \operatorname{argmin} \mathbb{E}[Y - g^{-1}(b_0 + b_1 X)]^2$. MLE$\to \beta^*$; sandwich s.e. valid (White 1982). Quasi-likelihood (Wedderburn 1974): correct two moments suffice.

5 The GLM framework

Three components: (i) exponential family $\mu = b'(\theta)$; (ii) $\eta = \mathbf{x}^\top \boldsymbol{\beta}$; (iii) link $g(\mu) = \eta$, canonical: $g = (b')^{-1}$.

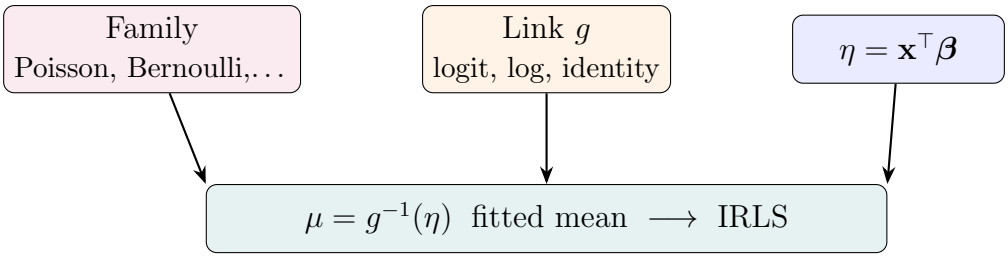
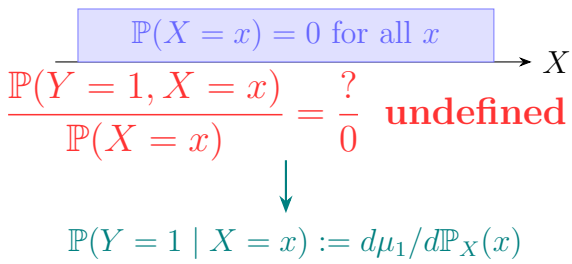


Table 1: Canonical links.

Family	Support	$b(\theta)$	Canonical $g(\mu)$	$V(\mu)$
Gaussian	$\mathbb{R}$	$\theta^2/2$	μ	1
Bernoulli	$\{0,1\}$	$\log(1+e^\theta)$	$\mathrm{logit}(\mu)$	$\mu(1-\mu)$
Poisson	$\mathbb{Z}_{\geq 0}$	e^θ	$\log \mu$	μ
Gamma	$(0,\infty)$	$-\log(-\theta)$	$-1/\mu$	μ^2
Inv. Gauss	$(0,\infty)$	$-\sqrt{-2\theta}$	$1/\mu^2$	μ^3

6 Logistic regression: Bernoulli case

6.1 Conditional probability for continuous X

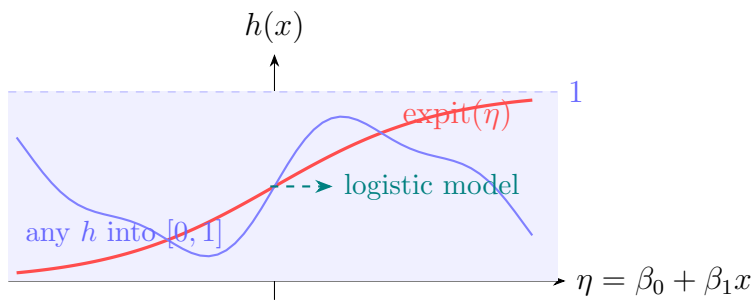


Definition 6.1 ($\mathbb{P}(Y = 1 \mid X = x)$ via R–N)**.** $\mu_1(B) := \mathbb{P}(Y = 1, X \in B)$. Then $\mu_1 \ll \mathbb{P}_X$ and $\mathbb{P}(Y = 1 \mid X = x) := d\mu_1/d\mathbb{P}_X(x)$. Recovers elementary ratio for discrete X.

Proposition 6.1. For $Y \in \{0,1\}$: $h(x) = \mathbb{P}(Y = 1 \mid X = x)$ $\mathbb{P}_X$-a.e.

Proof. Both are R–N derivatives of μ_1 w.r.t. $\mathbb{P}_X$. □

6.2 The logistic model



Range condition necessary but not sufficient. Logistic model restricts to the parametric family $\{\mathrm{expit}(\beta_0 + \beta_1 \cdot) : (\beta_0, \beta_1) \in \mathbb{R}^2\}$.

7 IRLS

Score equations: $\mathbf{X}^\top \mathbf{W} \mathbf{D}(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$, $W_{ii} = [V(\mu_i)(g'(\mu_i))^2]^{-1}$. Update:

$$\boldsymbol{\beta}^{(t+1)} = (\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{z}^{(t)}, \quad \mathbf{z}^{(t)} = \mathbf{X}\boldsymbol{\beta}^{(t)} + \mathbf{D}^{(t)}(\mathbf{y} - \boldsymbol{\mu}^{(t)}).$$

Logistic: $W_{ii} = \hat{p}_i(1 - \hat{p}_i)$. OLS: Gaussian, identity, $W = I$, one step.

8 Summary

1. **Atoms** (Lemma 1.1): $\sigma(X)$-measurable functions are constant a.e. on atoms $\{X = x_i\}$; this justifies $h(x_i) = \mathbb{E}[Y \mid X = x_i]$ as a single number.
2. **Full name**: R–N regression function of Y on X w.r.t. $\mathbb{P}$. Three-step construction (Remark 1.2). $\mathbb{P}$ need not be a probability measure.
3. **Degenerate parabola case** (degenerate case): $(\Omega, \mathcal{F}, \mathbb{P}) = ([0,1], \mathcal{B}([0,1]), \mathrm{Leb})$, $X = \mathrm{id}$, $Y = X^2$. Then $\mathbb{P} = \mathbb{P}_X$, $\{X \in B\} = B$, and the R–N construction gives $h = x^2$ without any averaging. The regression is trivial because Y is determined by X.
4. **Optimality proof** (optimality): $\mathbb{E}(Y - Z)^2 \leq \mathbb{E}(Y - U)^2$ for all $U \in \sigma(X)$. Key: decompose $E(Y - U)^2 = a + 2b + c$; the cross-term $b = 0$ because $\mathbb{E}[Y - Z \mid X] = 0$; remainder $c \geq 0$.
5. **Uniqueness**: h unique as $\mathbb{P}_X$-equivalence class, not pointwise.
6. R–N and C.E. definitions agree $\mathbb{P}_X$-a.e. (Thm. 1.2).
7. Conditional probability for continuous X requires R–N (Binary Y: $h = \mathbb{P}(Y = 1 \mid X = \cdot)$.
8. Canonical links derived via $g = (b')^{-1}$.

A Notation

$(\Omega, \mathcal{F}, \mathbb{P})$	probability space
$\mathbb{P}_X = X_* \mathbb{P}$	pushforward of X
$\mathrm{Range}(X) = X(\Omega)$	essential domain of h
$\nu(B) = \int_{\{X \in B\}} Y \, d\mathbb{P}$	signed measure
$h = d\nu / d\mathbb{P}_X$	R–N regression fn of Y on X w.r.t. $\mathbb{P}$
$\tilde{h}$	C.E. regression function
$Z = \mathbb{E}(Y \mid X)$	conditional expectation r.v.
$\mathbb{E}(Y \mid X = x) := h(x)$	notation only
$\ell^*(x)$	best affine approx. to h
$\mu_1(B) = \mathbb{P}(Y = 1, X \in B)$	Bernoulli measure
g	link function (fixed)
$b(\theta)$	log-partition function
$V(\mu)$	variance function
expit, logit	logistic and logit
IRLS	iteratively reweighted least squares
R–N	Radon–Nikodym